

Pet Health Tracker - Final Submission

Team: Total-H3 **Date:** December 5, 2025

1. Product Access Link

Live Application: <https://pet-health-tracker-hazel.vercel.app>

You can access the full application immediately at the link above. No installation required.

Demo Login Credentials:

- Email: `demo@pethealth.local`
- Password: `demo123456`

If you prefer to run the application locally, all instructions are in the GitHub repository below.

2. GitHub Repository Link

Repository: <https://github.com/jUNLINAAAA/pet-health-tracker>

The repository contains:

- **README.md** - Step-by-step instructions for launching and using the product
- **CODEBOOK.md** - Complete data dictionary with all variables, tables, and their descriptions
- **Source code** - Full Next.js application with TypeScript
- **Test suites** - Clinical validation tests and stress tests

Quick Start (from README)

```
# Clone the repository
git clone https://github.com/jUNLINAAAA/pet-health-tracker.git
cd pet-health-tracker

# Install dependencies
npm install

# Set up environment variables
cp .env.example .env.local
# Edit .env.local with your Supabase credentials

# Start development server
npm run dev

# Open browser
open http://localhost:3000
```

Latest Updates (December 2025)

- **Quick Insights API:** New `/api/quick-insights` backend endpoint for server-side aggregation of wellness index and alert statistics
- **Redesigned Quick Insights UI:** Sidebar widget now shows wellness index, pets count, and alert status (active/resolved/total) with proper loading states
- **Historical Charts:** Weight, activity, and score trends now display real backend data (60-day history)
- **Realtime Updates:** Dashboard syncs instantly via Supabase Realtime subscriptions

- **Comprehensive Alerts:** Rule-based alerts analyze ALL metrics (weight, activity, appetite, temperature, heart rate, clinical notes, vaccinations)
- **Auth Session Handling:** Fixed race condition ensuring data loads after authentication completes
- **Multi-User Support:** Full data isolation with Row Level Security (RLS) - each user sees only their own data
- **Sparklines:** Activity trend charts display proper 7-day time-series data with meaningful variations
- **Hydration Fix:** Prevents flash of unstyled content during hard refresh with proper loading states
- **Responsive Design:** Premium mobile-first UI with adaptive layouts for all screen sizes

Evaluation Criteria Response

Tool Implementation (60 pts)

Does it run? (30 pts)

Yes, the tool runs successfully.

We verified this by running the build process:

```
> npm run build

✓ Compiled successfully
✓ Generating static pages (31/31)
✓ Finalizing page optimization
```

The application is deployed and accessible at <https://pet-health-tracker-hazel.vercel.app>

What we built:

- A Next.js 13.5 web application with React 18 and TypeScript
- Connected to Supabase PostgreSQL database for data persistence
- Deployed on Vercel with automatic HTTPS and global CDN
- 63 unit tests passing (run with `npm test`)

Technology choices and why:

Technology	Purpose	Why We Chose It
Next.js 13.5	Framework	Server-side rendering for fast page loads, built-in API routes
Supabase	Database + Auth	Free tier, real-time subscriptions, PostgreSQL reliability
Vercel	Hosting	Zero-config deployment, automatic HTTPS, global edge network
TypeScript	Language	Catches bugs at compile time, better code documentation

Does it make sense? (30 pts)

Yes, the tool performs valid, evidence-based health analysis.

Our health scoring algorithm was validated against real clinical cases from peer-reviewed veterinary studies:

Clinical Validation Results:

- **93.8% accuracy** on 16 real veterinary cases
- **94.9% accuracy** on 1,323 parametric stress tests

How we validated:

We tested our algorithm against actual patient data from published veterinary research:

Study	What They Measured	How We Used It
German AJ et al. (2010)	Labrador obesity outcomes	Tested obese vs healthy Labs
Kealy RD et al. (2002)	14-year dog lifespan study	Validated lean vs overweight scoring
Cave NJ et al. (2012)	Cat obesity patterns	Tested DSH cats at different weights
Laflamme (1997)	Body Condition Score validation	Built our BCS-to-score mapping

Example of algorithm in action:

When a user enters a 42kg Labrador (breed ideal: 30kg), our algorithm:

- 1. Calculates deviation: 42kg is 40% above ideal
- 2. Maps to BCS 8 (obese) using Laflamme scale
- 3. Assigns weight score: 45/100
- 4. Generates insight: "Obese: 42kg (40% above ideal)"

This matches what a veterinarian would assess clinically.

The algorithm uses these evidence-based components:

Component	Weight	What It Measures	Source
Body Weight	20%	BCS score from weight vs breed ideal	Laflamme 1997
Activity Level	20%	Minutes of daily exercise vs targets	AAHA 2019 guidelines
Medical History	25%	Vaccinations, checkups, treatments	Standard veterinary practice
Active Alerts	15%	Current health concerns	Rule-based detection
Age Factor	10%	Life stage adjustments	AAHA/AAFP life stage guidelines
AI Insights	10%	Pattern detection from conversations	Context-aware analysis

Tool Design (30 pts)

Scope matches product development plan (10 pts)

Delivered all planned features:

Planned Feature	Delivered?	How It Works
Pet profiles	Yes	Create/edit pets with name, species, breed, age, weight, photo
Health tracking	Yes	Log weight, activity, temperature, appetite, heart rate, symptoms
Health scoring	Yes	Automated 7-component algorithm calculates overall health
AI document scanner	Yes	Upload vet documents, AI extracts health data automatically
AI health assistant	Yes	Chat with AI that knows your pet's health history
Alert system	Yes	Comprehensive algorithm-based alerts (see below)
Appointment scheduling	Yes	Track vet appointments with reminders

Alert System - All Metrics Analyzed:

The alert system uses a rule-based algorithm that analyzes ALL collected health metrics:

Metric	Alert Types Generated	Clinical Basis
Weight	obese, overweight, underweight, rapid_weight_gain, rapid_weight_loss	BCS scale (Laflamme 1997) + breed-specific ideals
Activity	severely_low_activity, low_activity, moderate_activity	AAHA exercise guidelines, age-adjusted
Appetite	poor_appetite, reduced_appetite, excessive_appetite	1-5 scale; abnormal = possible illness
Temperature	critical_fever, fever, hypothermia, low_temperature	Species-specific ranges (Dogs: 38-39.2°C)
Heart Rate	critical_tachycardia, elevated_heart_rate, critical_bradycardia, low_heart_rate	Species-specific (Dogs: 60-140 bpm, Cats: 140-220 bpm)
Vaccinations	vaccination_overdue, vaccination_due_soon, no_vaccination_record	Annual booster schedule
Age	senior_pet, geriatric_pet, impossible_age	Life expectancy by species/breed
Clinical Notes	urgent_clinical_attention, followup_reminder, medication_check, condition_monitoring	OCR-extracted vet records
Tracking	no_recent_tracking	Encourages consistent data logging

System integration (everything connects):

User adds health record → Triggers health score recalculation

- Checks for alert conditions
- Updates dashboard display (REALTIME)
- AI assistant gains new context

User uploads vet document → AI extracts data (vaccinations, weight, labs)

- Creates health records automatically
- Triggers alerts if abnormal results found

Real-time Updates:

- Dashboard metrics update instantly when data changes (via Supabase Realtime)
- Alert resolutions reflect immediately across all views
- Multi-device sync - changes on one device appear on others

Data Isolation & Security:

- Row Level Security (RLS) ensures each user only sees their own data
- No data mixing between accounts - complete privacy
- All tables protected: pets, alerts, appointments, health_records, health_scores, ai_health_insights

Clear user and use case (10 pts)

Primary Users:

1. **Pet owners** who want to track their pet's health over time
2. **Multi-pet households** managing several animals from one dashboard
3. **First-time pet parents** learning what's normal for their pet
4. **Caregivers of senior pets** who need to monitor health changes

Core Use Cases:

Situation	What User Does	What System Does
Daily check-in	Opens dashboard	Shows health score, alerts, upcoming appointments
Weekly weigh-in	Enters pet's weight	Calculates trend, updates score, alerts if concerning
After vet visit	Uploads document photo	AI extracts vaccinations, test results, creates records
Health question	Types question in chat	AI responds using pet's actual health data
Scheduling	Creates appointment	Saves to calendar, creates reminder alert

Why this matters:

Pet owners often don't realize their pet is gaining weight or missing vaccines until a vet visit. Our app provides continuous monitoring so issues are caught early.

Minimal user inputs (10 pts)

To get started, users only need to enter:

1. Pet name (text)
2. Species (dropdown: Dog, Cat, Bird, etc.)
3. Age (number)

That's it. Three fields to create a pet and start using the app.

Optional fields that improve accuracy:

- Breed (for more accurate weight targets)
- Current weight (for tracking)
- Photo (for identification)

What the system handles automatically:

- Health score calculation (no manual scoring)
- Alert generation (rule-based, not user-configured)
- Vaccine reminders (based on species and last recorded vaccine)
- Breed-specific targets (100+ breeds in database)

We deliberately avoided:

- Requiring users to know their pet's "ideal weight" (we calculate it from breed data)
 - Complex configuration screens (sensible defaults for everything)
 - Manual alert setup (alerts trigger automatically based on health data)
-

Tool Documentation (10 pts)

Fully reproducible materials

README.md includes:

- Project overview explaining what the app does
- Prerequisites (Node.js 18+, npm 9+)
- Step-by-step installation instructions with copy-paste commands
- Environment variable setup guide
- How to run the development server
- How to run tests
- Technology stack explanation

CODEBOOK.md includes:

Complete data dictionary for all 7 database tables:

Table	Purpose	Key Columns
pets	Pet profiles	id, name, species, breed, age, weight
health_records	Individual health entries	pet_id, type, value, unit, recorded_at
health_scores	Calculated scores	pet_id, overall, components, algorithm_version
alerts	Health warnings	pet_id, type, severity, message, is_resolved
appointments	Vet visits	pet_id, date_time, clinic_name, purpose
health_documents	Uploaded files	pet_id, file_name, extracted_data
assistant_messages	AI chat history	pet_id, role, content

Each table is documented with:

- Column name and data type
- Whether it's required or optional
- Description of what it stores
- Example values

Test suites included:

```
# Run unit tests (63 tests)
npm test

# Run clinical validation (16 real vet cases, 93.8% accuracy)
python3 tests/clinical-cases-test.py

# Run stress tests (1,323 parametric cases, 94.9% accuracy)
python3 tests/stress-test-algorithm.py
```

Summary

We built a pet health tracking application that:

1. **Runs correctly** - Builds successfully, deploys to Vercel, all features work
2. **Makes sense** - Uses evidence-based algorithm validated against real clinical data (93.8% accuracy)
3. **Has clear users** - Pet owners who want to monitor their animals' health
4. **Requires minimal input** - Just 3 fields to get started
5. **Is fully documented** - README with launch instructions, CODEBOOK with all data definitions

Links:

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